

Syllabus

MGT 100: Customer Analytics

Summer 2026

Welcome

We welcome everyone to this course. We want all students to feel valued, respected, and safe.

We also want you to succeed! We will work hard to help make that happen; we have good intentions toward you. However, this is not an easy class for most people, as we cover challenging material quickly. You will need to participate actively. This means engaging with readings, analytic tools, your instructors, and your peers. Rest assured that virtually all students pass this course and, we hope, learn a lot along the way.

This course was designed from scratch by UCSD faculty¹ for quantitative UCSD students. It serves as a core course in the joint Econ/Rady Business Economics major and as an alternate core course in the Business and Marketing minors. We actively work to maintain, update, and improve the course every time we teach it. We have received overwhelmingly positive comments about the value of the course, the style in which we've taught it, and the skills students gained as a result of studying this material. We endeavor to provide you the same challenging yet enjoyable experience.

Instructors

Prof. Dan Yavorsky

- email: dyavorsky@ucsd.edu
- cell: 951-201-0927
- websites: danyavorsky.com | [linkedin.com/in/dyavorsky](https://www.linkedin.com/in/dyavorsky)
- office hours: over zoom, just email me to schedule

Teaching Assistants

- **Heather Li**: heather.li@rady.ucsd.edu
- **Merna Saad**: merna.saad@rady.ucsd.edu
- No formal TA Sessions, but available over zoom, email to schedule

Logistics

Class is **11:00a-2:00p** via Zoom on **Wednesdays** and **Fridays**. I will aim to record and post lectures to Canvas within 24 hours after class.

All materials for this course — including books, articles, videos, blogs, and visualizations, as well as slides, R code scripts, and datasets — are **free** and available on (or linked from) Canvas.

The primary method of Q&A will be **Piazza**.

¹This course was co-designed with the amazing UCSD Professor Ken Wilbur. Materials for his version of the course are available online at <https://kennethcwilbur.github.io/mgt100/>.

Course Introduction

Customer Analytics is the use of customer data — often combined with domain knowledge, relevant theory, and statistical modeling — to inform and improve business decision-making. Our primary goals are to develop student understanding of data-driven business decision making, and to enable students to perform and interpret analytic techniques whose results inform those decisions.

In pursuit of those goals, our course design principles are *experiential* learning and assessment of applications. Half our time will be spent discussing key concepts in customer analytics. The other half will be spent implementing those ideas by conducting analysis. Our mentors instilled in us the idea that “you don’t understand it until you code it,” and we, in turn, aim to propagate this belief with the next generation of scholars. Implementation will all be done in the R programming language.

Class meetings will have a regular format. Each class session we will include a lecture followed by an analytic (i.e., coding or programming) demonstration. During the demonstration, we will step through an R code script to implement techniques from that session’s lecture.

Outside of the classroom, students will complete required readings and homework assignments consisting of a set of analytic tasks similar to what was demonstrated in class. There is no midterm; there is a final exam.

This is designed as a survey course: we cover a broad range of topics in limited depth, although we maintain a deeper through-line that investigates demand modeling and usage. The survey nature of the course is more typical of graduate business courses than the undergraduate economics courses many students may have taken previously.

We seek to simulate a professional experience. We therefore expect consistent, regular attendance and participation. We require no memorization, encourage collaboration, and will aim to provide sufficient time and resources to complete deliverables.

We understand that student financial resources are often limited. We rely exclusively on materials that are either free or already paid by your tuition. We then provide pointers to additional or advanced material for students interested in deeper learning.

Most students will need to commit 8–10 hours per week outside of class to have a successful experience. Student feedback is welcome at any point.

Topics and Schedule

Please see the associated Canvas modules to find related materials and more information.

#	Day	Date	Topic	Due
1	W	8/05	Introduction to Course and R	
2	F	8/07	Customer Data and Data Visualization	HW 1: Mon 8/10
3	W	8/12	Cluster Analysis and Market Segmentation	
4	F	8/14	Dimension Reduction and Market Mapping	HW 2: Mon 8/17
5	W	8/19	The Multinomial Logit Model and Demand Estimation	
6	F	8/21	Conjoint Analysis and Heterogeneity	HW 3: Mon 8/24
7	W	8/26	Pricing and Price Optimization	
8	F	8/28	Branding	HW 4: Mon 8/31
9	W	9/02	Market Sizing and Customer Lifetime Value	
Final Exam: F 9/04 through Sa 9/05				

Homework is due at **11:59pm** on the Monday listed.

Assignments and Grading

Your work in this course is scored out of a base of **100 points**, and bonus points are then added *on top* of your base score. There is no fixed maximum, and strong students finish above 100.

Component	Base	Bonus available
Homeworks (4 assignments $\times$ 10 points)	40	up to +40
Final exam (4 parts $\times$ 10 points)	40	up to +20
Engagement	20	up to +5
Total	100	

How your work is scored: Each homework, and each part of the final exam, is worth up to 10 points. Because you are encouraged to use AI, a correct answer is not by itself distinguishing — we expect most submissions to be largely correct. What we score is therefore what *you* added on top of what a tool would produce:

- **10 points** — the submission goes materially beyond the analysis demonstrated in class. It interprets the result for the decision at hand, questions an assumption, checks something nobody asked you to check, or explains something surprising in the output. In short, this is the score for work that was going for the bonus and simply did not happen to win it.
- **8 points** — correct and complete, with evidence of independent thought, such as a genuine interpretation or a well-chosen caveat, but not much beyond what was demonstrated in class.
- **5 points** — submitted and technically adequate, but showing little sign of engagement: generic interpretation, output presented without examination, boilerplate caveats.
- **0 points** — not submitted.

Note that we score the substance of what you contributed, not our guess about which tools you used. Using AI is permitted and encouraged; leaning on it without adding anything of your own will cost you points.

What earns a bonus: Bonuses reward the same qualities, taken further: the depth of your interpretation, the judgment behind your recommendation, analysis that goes well beyond what the question asked, clear exposition and well-designed figures, and honest treatment of the assumptions and limitations behind your results.

Homeworks (40 points): There are 4 homework assignments, following each week of the course (except the last). On each assignment, the five best submissions in the class will be awarded 10 bonus points, and the next five best will be awarded 5 bonus points.

Final Exam (40 points): There will be an online final exam. The structure will be similar to that of a homework assignment, and it will consist of 4 parts. Separately, for each part, the ten best submissions will be awarded 5 bonus points.

Engagement (20 points): This part of your grade assesses your participation and contribution. It can be met either by regular attendance via zoom, by watching the lecture recordings and suggesting enhancements to the slides, and/or by contributing to productive Q&A on Piazza. Specifically:

- **Attendance**, worth 3 points per lecture attended via zoom, up to a maximum of 20 points. Because there are 9 lectures, attending **7 of the 9** earns full attendance credit: you may miss two sessions without needing to explain them or make them up.
- **Slide enhancements**, worth 2 points each, up to a maximum of 12 points. If you are unable to make the lectures due to other commitments or an incompatible time zone difference, you may watch the recordings and suggest enhancements to the slides instead. Enhancements may include (but are not limited to) adding an on-point example, re-doing a slide to make the exposition more clear, providing an animation to replace one of my images, or suggesting good books or academic articles to support a point made. I am looking for a small number of substantive contributions rather than a long list of minor ones.
- **Piazza contributions**, worth 1 point per relevant question and 1 point per thoughtful answer, up to a maximum of 10 points.

You may combine these methods. Engagement contributes at most 25 points in total: the first 20 count toward your base score, and anything above 20 is bonus. Attendance alone can reach the full 20, whereas enhancements and Piazza together reach 22, so regular attendance is the most direct route to full engagement credit. The enhancement path is meant to supplement attendance, or to substitute for it when a genuine conflict makes attending impossible.

Grade Calculations: Letter grade cutoffs are *not* set in advance. We total everyone's points at the end of the course, look at the resulting distribution, and place the cutoffs where they fall most fairly. Do not assume any particular numeric threshold corresponds to a particular letter grade. The median grade will be curved to a B or B+, and in a typical offering of this course roughly 45% of students earn a grade in the A range. If more than 80% of SET evaluations are completed, some bonus will be applied.

Course Policies

Collaboration: All assignments except the final exam may be worked on in collaboration with other students. Collaboration is optional and groups should be small. Each student is individually responsible for creating their own work and submitting their own answers or results.

Contacting Instructors: Use Piazza to ask questions about course content and homeworks publicly so that both the instructors and other students can provide answers. For individual matters (illness, questions about grading, etc.) please email the instructor and cc the TA(s) (or vice-versa); do not email us separately.

FinAid Survey: In the last question of the FinAid Survey during week 1, you can receive extra credit for this course by submitting “pitbull” along with any additional answer to the question. Please do not inform your fellow students of this.

When Struggling: We understand that student learning styles differ and no single approach is best for everyone. We also know that anyone can go through a difficult time. Please tell us if you have trouble learning in this environment. We may be able to make suggestions, connect you with resources, or find appropriate accommodations. We will work with you as best we can.

Use of AI Technology: We explicitly allow and encourage use of AI technology on all coursework, including the final exam. We caution that you are responsible for content and accuracy of your submitted work.

Class Recordings: Class sessions will be recorded and posted to Canvas for the use of students enrolled in this course. If you speak or turn on your camera, your name, voice, and image may be captured in the recording; if you would rather not appear, you are welcome to keep your camera off and to participate through Piazza instead.

Late Submissions: Late deliverables will be accepted for partial credit; full credit may only be obtained in grave circumstances with some form of documentation provided prior to the deliverable due date.

Re-grade Requests: Any request for regrading must be made in writing within two weeks of a deliverable being assessed but before final course grades are submitted to the Registrar. The professor and/or TA(s) will entirely regrade the deliverable such that the resulting grade change may be positive or negative.

Important UCSD Topics

We adopt the following policies based on university guidance.

Academic Integrity

Academic Integrity is expected of everyone at UC San Diego. This means that you must be honest, fair, responsible, respectful, and trustworthy in all of your actions. Lying, cheating, or any other forms of dishonesty will not be tolerated because they undermine learning and the University's ability to certify students' knowledge and abilities. Thus, any attempt to get, or help another get, a grade by cheating, lying, or dishonesty will be reported to the Academic Integrity Office and will result in sanctions. Sanctions can include an F in this class and suspension or dismissal from the University.

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

You can learn more about academic integrity at:

<https://academicintegrity.ucsd.edu/>

The complete UCSD Policy on Integrity of Scholarship can be viewed at:

<http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

All aspects of the UCSD honor code apply in this course. If you are ever unsure how they apply, please ask your classmates, TA, or professor for clarification. It is much better to be conservative about honor code violations than to take a risk. You can be suspended or expelled for cheating.

Students with Disabilities

A student who has a disability or special needs and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the instructor and OSD liaison.

Students requesting accommodations for this course due to a disability must provide a current Authorization for Accommodation (AFA) letter (paper or electronic) issued by the OSD. Students are required to discuss accommodation arrangements with instructors and OSD liaisons in the department 72 business hours in advance of any exams or assignments. No accommodations can be implemented retroactively.

Please visit the OSD website <https://osd.ucsd.edu/portal/tutorial.html> for further information or contact the Office for Students with Disabilities by phone at 858-534-4382 or via email at osd@ucsd.edu.

NonDiscrimination Policy Statement

The University of California, in accordance with applicable Federal and State law and University policy, does not discriminate on the basis of race, color, national origin, religion, sex, gender identity, pregnancy, physical or mental disability, medical condition (cancer-related or genetic characteristics), ancestry, marital status, age, sexual orientation, citizenship, or service in the uniformed services.

The University also prohibits sexual harassment. This nondiscrimination policy covers admission, access, and treatment in University programs and activities.

Title IX

The Office for the Prevention of Harassment & Discrimination (OPHD) provides assistance to students, faculty, and staff regarding reports of bias, harassment, and discrimination. OPHD is the UC San Diego Title IX office. Title IX of the Education Amendments of 1972 is the federal law that prohibits sex discrimination in educational institutions that are recipients of federal funds. Rady students have the right to an educational environment that is free from harassment and discrimination.

You can make a complaint of harassment or discrimination — or simply make an appointment to find out more information — by contacting OPHD:

- by phone at 858-534-8298
- by email at ophd@ucsd.edu
- or online at the Overview for Students webpage

Students may feel more comfortable discussing their particular concern with a trusted employee. This may be a Rady student affairs staff member, a department Chair, a faculty member, or other University official. These individuals have an obligation to report incidents of sexual violence and sexual harassment to OPHD. This does not necessarily mean that a formal complaint will be filed.

If you find yourself in an uncomfortable situation, ask for help. The Rady School of Management is committed to upholding University policies regarding nondiscrimination, sexual violence, and sexual harassment.

Health and Well-Being

Throughout your time at UC San Diego, you may experience a range of issues that can negatively impact your learning. These may include physical illness, housing or food insecurity, strained relationships, loss of motivation, depression, anxiety, high levels of stress, alcohol and drug problems, feeling down, interpersonal or sexual violence, or grief.

These concerns or stressful events may lead to diminished academic performance and affect your ability to participate in day-to-day activities. If there are issues related to coursework that are a source of particular stress or challenge, please speak with your professors so that we are able to support you. In addition, UC San Diego provides a number of resources to all enrolled students, including:

- Counseling and Psychological Services: 858-534-3755 or caps.ucsd.edu
- Student Health Services: 858-534-3300 or studenthealth.ucsd.edu
- CARE at the Sexual Assault Resource Center: 858-534-5793 or care.ucsd.edu
- The Hub Basic Needs Center: 858-246-2632 or basicneeds.ucsd.edu

For example, if you are skipping and stretching meals, or having difficulties affording or accessing food, you may be eligible for CalFresh, California's Supplemental Nutrition Assistance Program, that can provide up to \$298 a month in free money on a debit card to buy food. Students can apply at benefitscal.com/r/ucsandiegocalfresh. Relatedly, the Hub Basic Needs Center empowers all students by connecting them to resources for food, stable housing, and financial literacy.